

1 · INDICATIONS

Diagnostic	Effusion of unknown etiology
Therapeutic	Large effusion w/ dyspnea/hypoxemia; effusion at risk of complication if not drained (parapneumonic)

2 · CONTRAINDICATIONS

Hard stops	Overlying skin infection; <2 cm fluid; isolated multiloculated effusion needing chest tube (dx tap may still be OK)
Cautions (not absolute)	Full anticoag / INR >2 / plt <50K (bleed); mech vent (PTX); lung entrapment (RPE)

Coags rarely block the tap: abnormal INR/plt alone NOT linked to more bleeding after US-guided thora (Patel AJR 2011; Ault Thorax 2015). Don't delay a needed dx tap in sepsis for a mild INR.

3 · US SITE SELECTION

- **Low-freq curvilinear** (depth >8 cm): map pocket + 4 landmarks (chest wall, parietal pleura, lung, diaphragm, liver/spleen).
- **High-freq linear + color Doppler:** screen aberrant subcostal vessel; measure skin → pleura.

Rule	Why
Pocket ≥2 cm	Below 2 cm risks hitting lung
OVER upper rib margin	NV bundle runs UNDER the rib
>9–10 cm lateral to spine	Vessels most tortuous medially
≥1 space above diaphragm	Avoid liver/spleen injury

4 · TECHNIQUE

- **Time-out** (name/DOB/site/site/allergies) before opening sterile supplies.
- **Prep:** chlorhexidine **scrub** ×30 s (iodine = apply & dry); fenestrated drape.
- **Anesthesia:** 1% lido wheal → advance under negative pressure to fluid → ~2 cc just superficial to pleura.
- **#11 scalpel nick** → catheter-over-needle OVER the rib w/ negative pressure; on fluid, advance 0.5–1 cm, thread catheter to hub, remove needle.

SAFETY: go over the TOP of the rib; get the **CATHETER (not needle)** into the space — needle-in-chest drainage causes pneumothorax. Hit rib → lift over it, don't pivot.

5 · DRAIN & STOP RULES

- IV tubing w/ two one-way valves; pull-to-fill, push-to-empty.
- **STOP for:** chest pain, dyspnea, severe cough; resistance / pleural pressure < **−20 cm H₂O**; or no flow.

No fixed volume: the <1.5 L "limit" is arbitrary & not evidence-based — drain to **symptoms/pressure**. RPE <1%; routine manometry doesn't clearly cut RPE (Lentz 2019).

6 · FLUID STUDIES

Suspect	Send (volume)
All	Cell count+diff, LDH, protein (≥10 cc) — apply Light's
Infection	Gram stain + cx (≥20 cc); pH on ice
Malignancy	Cytology (≥30–60 cc)
Chylothorax	Triglycerides, cholesterol

Complicated parapneumonic / empyema = pH <7.2, low glucose, +Gram stain/pus → **CHEST TUBE**, not abx alone (BTS 2023).

LIGHT'S CRITERIA (exudate if ≥1)

- Pleural/serum **protein** >0.5
- Pleural/serum **LDH** >0.6
- Pleural **LDH** >¾ ULN of serum LDH

Closure: stopcock to patient → brisk catheter removal → pressure >1 min → dressing. Concern for PTX → lung US (anterior, supine) or CXR.

MASTERY-LEARNING THRESHOLDS

Component	MPS (sim)	UPS (clinical)
Mistakes (Angoff)	<24	<7
Global skills	>2	≥proficient
Entrustment	>2	indirect supervision